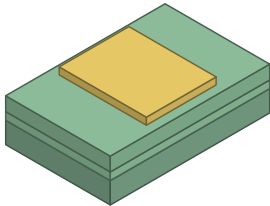


III-V and III-nitride light emitters

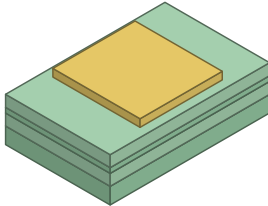
From homojunctions to heterostructures, resonators, and mesas

(a) GaAs junction laser



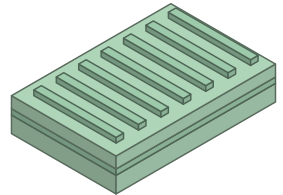
single-material p-n gain
Hall et al., PRL 9 (1962)

(b) AlGaAs/GaAs DH laser



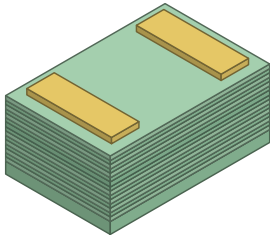
carrier + optical confinement
Hayashi et al., APL 17 (1970)

(c) InP/InGaAsP DFB laser



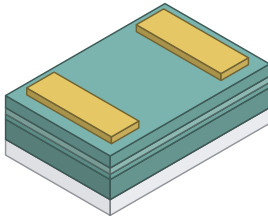
buried active layer + grating
Kogelnik & Shank, APL 18 (1971)

(d) GaAs/AlGaAs VCSEL



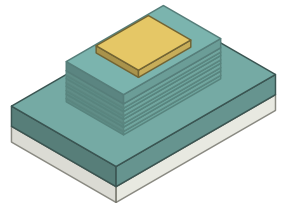
repeated DBRs + vertical cavity
Soda et al., JJAP 18 (1979)

(e) InGaN/GaN blue LED



on sapphire, DH + blocking layer
Nakamura et al., APL 64 (1994)

(f) InGaN/GaN micro-LED



on SiC, etched mesa + MQWs
Jiang et al., APL 78 (2001)